

Algorithm

An algorithm is a step-by-step set of instructions for solving a problem or performing a task.

You can think of it like a recipe: just like a recipe tells you how to make a dish in a specific order, an algorithm tells a computer (or a person) how to do something from start to finish.

Algorithms are used **everywhere**, especially in tech and daily life.

Awesome! Let's take a real-world example: **an algorithm for making a cup of tea** ☕

Here's the step-by-step breakdown (the algorithm):

1. Boil water.
2. Place a teabag in a cup.
3. Pour the hot water into the cup.
4. Let the tea steep for 3–5 minutes.
5. Remove the teabag.
6. Add sugar or milk if desired.
7. Stir.
8. Serve and enjoy!

Algorithm to add 3 numbers and print their sum:

1. Start
2. Declare 3 integer variables num1, num2, and num3.
3. Take the three numbers, to be added, as inputs in variables num1, num2, and num3 respectively.

4. Declare an integer variable sum to store the resultant sum of the 3 numbers.
5. Add the 3 numbers and store the result in the variable sum.
6. Print the value of the variable sum
7. End

Code

```
# Initialize 3 variables to zero
num1 = num2 = num3 = 0

# Variable to store the resultant sum
result = 0

# Take the 3 numbers as input
num1 = int(input("Enter the 1st number: "))
num2 = int(input("Enter the 2nd number: "))
num3 = int(input("Enter the 3rd number: "))

# Calculate sum using + operator and store in variable result
result = num1 + num2 + num3

# Print the sum
print("Sum of the 3 numbers is:", result)
```

output:

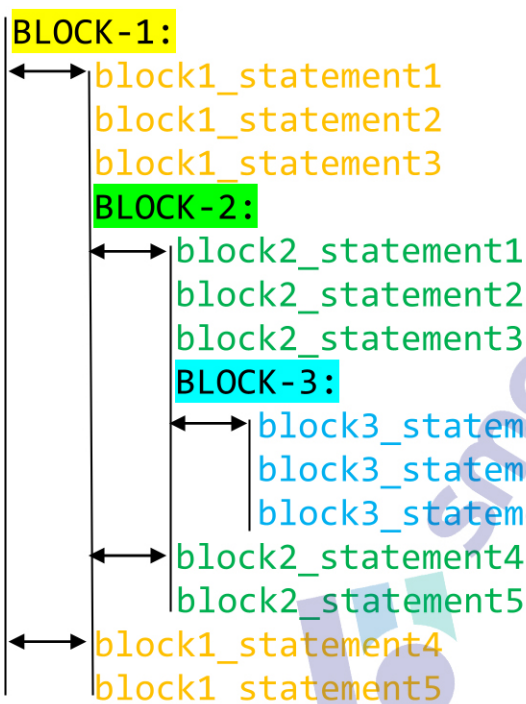
```
Enter the 1st number: 10
Enter the 2nd number: 20
Enter the 3rd number: 30
Sum of the 3 numbers is: 60
```

Python indentation

Python indentation uses to define the block of the code. The other programming languages such as C, C++, and Java use curly braces {}, whereas Python uses an indentation. Whitespaces are used as indentation in Python.

Indentation uses at the beginning of the code and ends with the unintended line. That same line indentation defines the block of the code (body of a function, loop, if, class etc.)

Generally, four whitespaces are used as the indentation. The amount of indentation depends on user, but it must be consistent throughout that block.



Control Statements in Python

Decision making is the most important aspect of almost all the programming languages. As the name implies, decision making allows us to run a particular block of code for a particular decision. Here, the decisions are made on the validity of particular conditions. Condition checking is the backbone of decision making.

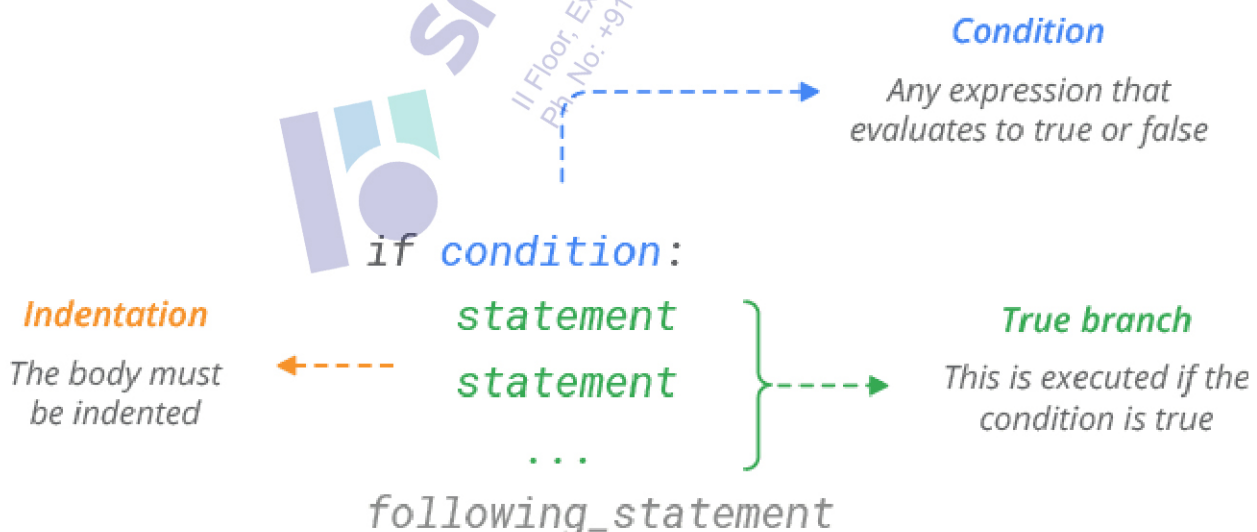
Conditional Statements:

Simple if statement

The, if statement is used to test a particular condition and if the condition is true, it executes a block of code known as if-block. The condition of if statement can be any valid logical expression which can be either evaluated to True or False.

The syntax of the if-statement is given below.

```
if(expression):  
    #statement
```



Condition is True

```
number = 10  
if number > 0:  
    # code
```

Q. Check whether a number is even

```
num = int(input("Enter the number:"))  
if(num%2==0):  
    print("Number is even")
```

output:

enter the number:12

Number is even

Q. Program to print the largest of the three numbers.

```
a=int(input("Enter the value for a:"))  
b=int(input("Enter the value for b:"))  
c=int(input("Enter the value for c:"))  
if(a>b and a>c):  
    print("a is largest")  
if(b>a and b>c):  
    print("b is largest")  
if(c>a and c>b):  
    print("c is largest")
```

output:

Enter the value for a:10

Enter the value for b:20

Enter the value for c:30

c is largest

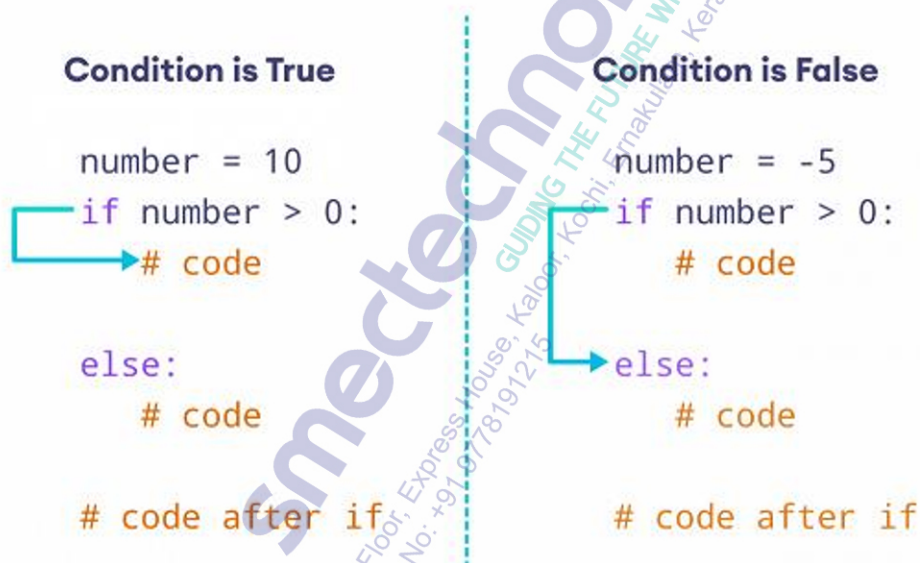
The if-else statement

The if-else statement provides an else block combined with the if statement which is executed in the false case of the condition.

If the condition is true, then the if-block is executed. Otherwise, the else-block is executed.

The syntax of the if-else statement is given below.

```
if(condition):  
    #block of statements  
else:  
    #another block of statements (else-block)
```



Q. Find largest of two numbers

```
a=int(input("Enter a:"))  
b=int(input("Enter b:"))  
  
if(a>b):  
    print("a is largest")  
else:  
    print("b is largest")
```

output:

Enter a:10

Enter b:20

b is largest

Q. Program to check whether a person is eligible to vote or not.

```
age = int (input("Enter your age:"))  
  
if(age>=18):  
    print("You are eligible to vote !!")  
else:  
    print("Sorry! you have to wait !!")
```

output:

Enter your age:19

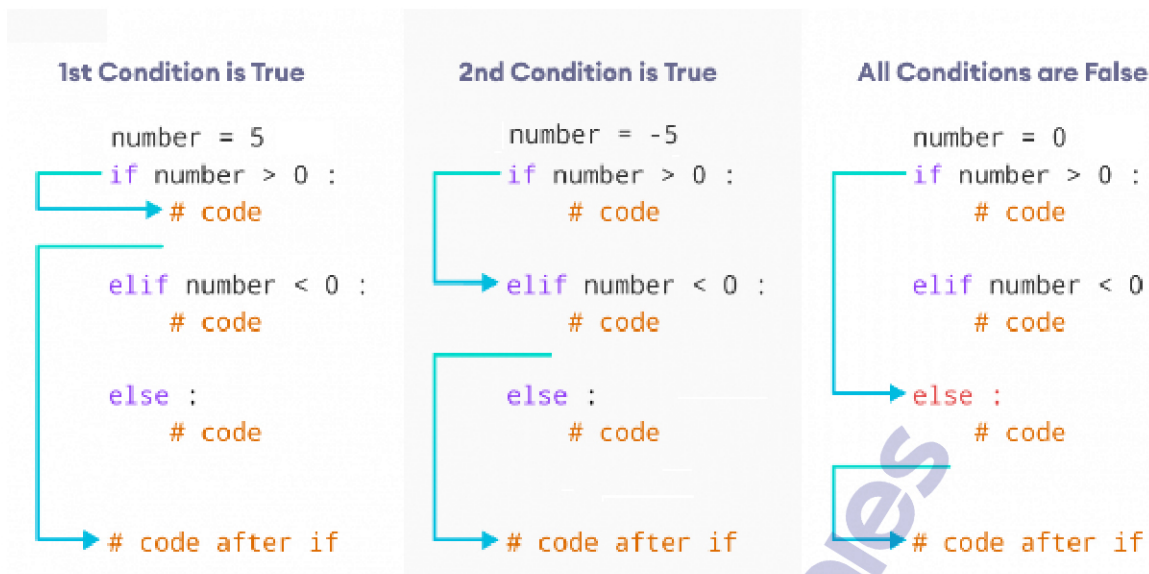
You are eligible to vote!!

The elif statement

The elif statement enables us to check multiple conditions and execute the specific block of statements depending upon the true condition among them. We can have any number of elif statements in our program depending upon our need.

The syntax of the elif statement is given below.

```
if(expression 1):  
    # block of statements  
elif(expression 2):  
    # block of statements  
elif(expression 3):  
    # block of statements  
else:  
    # block of statements
```



Q. Example to check the number entered

```
number = int(input("Enter the number:"))
if(number==10):
    print("number is equals to 10")
elif(number==50):
    print("number is equal to 50")
elif(number==100):
    print("number is equal to 100")
else:
    print("number is not equal to 10, 50 or 100")
```

output:

Enter the number:50
number is equal to 50

Q.To print the day of the week by inputting day number

```
day = int(input("Enter the number:"))
if(day==1):
    print("Sunday")
if(day == 2):
    print("Monday")
if(day == 3):
    print("Tuesday")
if(day == 4):
    print("Wednesday")
if(day == 5):
    print("Thursday")
if(day == 6):
    print("Friday")
if(day == 7):
    print("Saturday")
```

output:

Enter the number:4

Wednesday

Q. Student grade calculation

```
marks = int(input("Enter the marks:"))
if(marks > 85 and marks <= 100):
    print("Congrats ! you scored grade A ...")
elif(marks > 60 and marks <= 85):
    print("You scored grade B + ...")
elif(marks > 40 and marks <= 60):
    print("You scored grade B ...")
elif(marks > 30 and marks <= 40):
    print("You scored grade C ...")
else:
    print("Sorry you are fail ?")
```

output:

Enter the mark:70

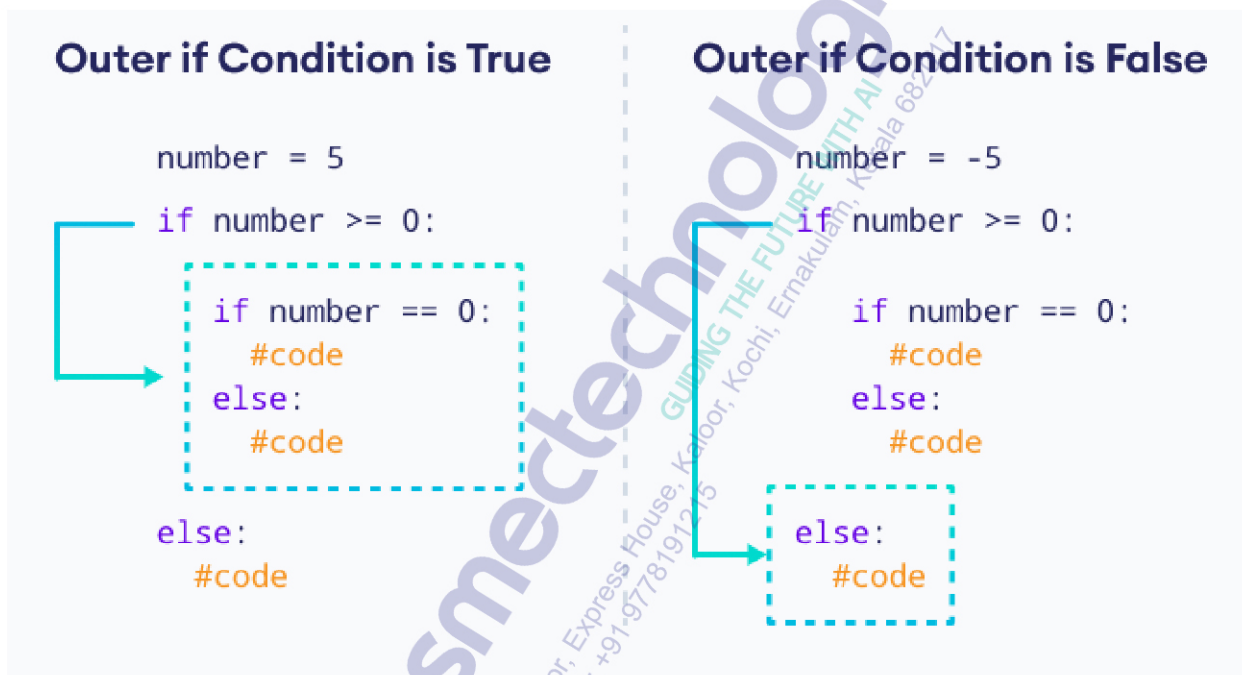
You scored grade B + ...

Nested-if

Python allows us to nest if statements within if statements. i.e, we can place an if statement inside another if statement.

Syntax:

```
if(condition1):  
    # Executes when condition1 is true  
    if(condition2):  
        # Executes when condition2 is true  
    # if Block is end here  
# if Block is end here
```



Q. Find largest of 3 numbers using nested if

```
a=int(input("Enter a:"))  
b=int(input("Enter b:"))  
c=int(input("Enter c:"))  
if(a>b):  
    if(a>c):  
        print("a is Larger")  
    else:
```

```
        print("c is Larger")
else:
    if(b>c):
        print("b is Larger")
    else:
        print("c is Larger")
```

output:

Enter a:10

Enter b:20

Enter c:30

c is Larger

match case statement

A Python match-case statement takes an expression and compares its value to successive patterns given as one or more case blocks. Only the first pattern that matches gets executed. It is also possible to extract components (sequence elements or object attributes) from the value into variables.

Syntax:

```
match(variable_name):
```

```
    case 'pattern 1' :
```

```
        statement 1
```

```
    case 'pattern 2' :
```

```
        statement 2
```

```
    ...
```

```
    case 'pattern n' :
```

```
        statement n
```

Q. To print day of the week by inputting day number using match statement

```
num=int(input('Enter the number:'))
```

```
match(num):  
    case 0:  
        print("Monday")  
    case 1:  
        print("Tuesday")  
    case 2:  
        print("Wednesday")  
    case 3:  
        print("Thursday")  
    case 4:  
        print("Friday")  
    case 5:  
        print("Saturday")  
    case 6:  
        print("Sunday")  
    case _:  
        print("Invalid day number")
```

output:

Enter the number:3

Thursday



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